Lesson Plan

Python Text Mining and Analytics: Transforming Text into Insights

TGS-2026064715 · WSQ · 2 days / 16 hours · v15 · 14 September 2026

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Document Version Control Record

Version	Release date	Change	Owner
v14	Previous release	Legacy courseware	Tertiary Infotech Academy
v15	14 September 2026	Corrected course code; rebuilt visual slides; aligned synthetic-data labs, LG and LP	Tertiary Infotech Academy

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Course schedule

16 hours total including 2.5 hours assessment: 7.5 hours classroom facilitation, 6 hours practical learning, 1 hour WA and 1.5 hours PP, following the approved course proposal.

Day 1 — 8 hours

Topic 1: CRISP-DM and corpus design

Slides 4–47. Trainer demonstrates the evidence trace and checks learner interpretation.

- Lab 01: lab-01-business-question. Learners run the script, inspect output, and discuss the failure mode.
- Lab 02: lab-02-corpus-ingestion. Learners run the script, inspect output, and discuss the failure mode.

Topic 2: Text cleaning and vectorization

Slides 49–88. Trainer demonstrates the evidence trace and checks learner interpretation.

- Lab 03: lab-03-clean-tokenize. Learners run the script, inspect output, and discuss the failure mode.
- Lab 04: lab-04-stemming-lemmatization. Learners run the script, inspect output, and discuss the failure mode.

Topic 3: Text analytics and feature engineering

Slides 90–129. Trainer demonstrates the evidence trace and checks learner interpretation.

- Lab 05: lab-05-tf-ngram. Learners run the script, inspect output, and discuss the failure mode.
- Lab 06: lab-06-tfidf. Learners run the script, inspect output, and discuss the failure mode.

Attendance, feedback and learning checks are recorded in the LMS/TMS.

Day 2 — 8 hours

Topic 4: Sentiment modelling and evaluation

Slides 131–174. Trainer demonstrates the evidence trace and checks learner interpretation.

- Lab 07: lab-07-pos-ner-links. Learners run the script, inspect output, and discuss the failure mode.
- Lab 08: lab-08-cluster-classify. Learners run the script, inspect output, and discuss the failure mode.

Topic 5: Summarization and visualization

Slides 176–223. Trainer demonstrates the evidence trace and checks learner interpretation.

- Lab 09: lab-09-sentiment-model. Learners run the script, inspect output, and discuss the failure mode.

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- Lab 10: lab-10-summarize-visualize. Learners run the script, inspect output, and discuss the failure mode.

Attendance, feedback and learning checks are recorded in the LMS/TMS.

Assessment

WA (SAQ): 60 minutes, K1–K12. PP: 90 minutes, A1–A9. Both are individual, open book, and submitted in the LMS. Provide briefing before assessment; collect and sign off after.

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